

TILE STATION

Complete System Overview

EPOS | E-Commerce Website | Admin Back Office
Supplier Management | Stock Control | Trade Accounts

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Table of Contents

1. System Overview
2. E-Commerce Website (Customer-Facing)
3. EPOS - Point of Sale
4. Admin Back Office
5. Supplier Management & Sync Hub
6. Stock Management
7. Sales & Documents
8. Customer & Trade Accounts
9. Communication Hub
10. Reports & Analytics
11. Website Management
12. Security & User Management

1. System Overview

Tile Station runs on a fully integrated platform covering three core areas: a customer-facing e-commerce website, an in-store EPOS (Electronic Point of Sale) system, and a comprehensive admin back office. All three share the same product database, stock levels, customer records, and pricing engine - ensuring consistency across online and in-store operations.

Technology

- Frontend: React web application (responsive desktop & mobile)
- Backend: FastAPI (Python) REST API
- Database: MongoDB (single source of truth)
- Hosting: Railway (production), Emergent (preview/staging)
- 3 Showroom Locations: Tonbridge, Gravesend, Chingford

Key Integrations

- WhatsApp Cloud API - automated trade welcome messages & order notifications
- AI Description Generator - GPT-powered product descriptions
- Barcode Scanner - scan-to-sell in showrooms
- Supplier Scraping - automated price/stock sync from supplier websites

2. E-Commerce Website (Customer-Facing)

The public-facing website at tilestation.co.uk allows retail and trade customers to browse, configure, and purchase tiles, flooring, bathroom products, materials, and accessories online.

Product Browsing

- Product Groups: All Tiles, Flooring, Bathroom, Underfloor Heating, Materials, Tools & Accessories
- Collection-based browsing - products grouped by series/collection with variant swatches
- Category tabs within each group (e.g. Flooring > Engineered Wood, Laminate, SPC/LVT)
- Advanced filtering: colour, finish, size, material, price range, room suitability, slip rating, stock
- Multiple view modes: list, grid (3-col), grid (4-col)
- Sort by: price, name, popularity, newest

Product Detail Pages

- Colour/Variant selector with product images
- Finish selector - only shows finishes available for selected colour
- Size selector with per-size pricing
- Volume pricing table - buy more, save more (dynamic tier discounts)
- Tile Calculator - enter room dimensions, auto-calculates m2 + boxes needed
- Sale pricing with WAS/NOW and percentage savings
- Stock availability indicator
- Made in [Country] badge with flag
- About This Product - AI-generated or admin-written descriptions
- Image gallery with zoom

Shopping & Checkout

- Shopping cart with box/m2 quantity modes
- Sample ordering service (free tile samples)
- Wishlist functionality
- Guest checkout or account-based checkout
- Order tracking page
- Recently viewed products

Trade Customer Features

- Trade account registration & approval workflow
- Trade login with exclusive ex-VAT pricing
- Trade volume pricing (compounded: sale + trade + volume discounts)
- Credit back display on trade orders
- Trade-only pricing labels and badges

Other Pages

- New Collections showcase page
- Clearance/Sale page
- Bathroom visualiser page
- Room Visualiser
- Store locations page with showroom details
- Contact page

- Delivery information, Returns & Refunds, FAQs, Privacy Policy, T&Cs

3. EPOS - Point of Sale

The EPOS system is used in all three Tile Station showrooms (Tonbridge, Gravesend, Chingford) for walk-in sales, quotations, and stock lookups. It shares the same product database as the website.

Core EPOS Features

- Barcode scanner integration - scan product barcodes to add to sale
- Product search by name, SKU, or barcode
- Real-time stock levels across all showrooms
- Cash counter / till management
- Staff PIN login - quick switch between staff members
- Showroom-specific EPOS views per location

Sales Documents (from EPOS)

- Invoices - generate, print, email to customer
- Quotations - create quotes with configurable validity period
- Cash Quotations - quick quotes for walk-in customers
- Proforma Invoices - pre-payment invoices
- Credit Notes - issue credits for returns/adjustments
- Refunds - process full or partial refunds

Showroom Management

- Multi-showroom support (Tonbridge, Gravesend, Chingford)
- Showroom-specific stock allocation
- Stock transfers between showrooms
- Delivery check-in - log incoming deliveries from suppliers
- Price tickets - generate shelf labels with barcodes

4. Admin Back Office

The admin panel is the command centre for managing all aspects of the business - products, pricing, orders, customers, stock, suppliers, and the website.

Dashboard

- Sales overview - today, this week, this month
- Recent orders and pending actions
- Stock alerts and low-stock warnings
- Quick links to all admin sections

Admin Sections

- Sales & EPOS Hub - invoices, quotations, refunds, credit notes, cash counter
- Products & Suppliers Hub - supplier products, sync hub, product editor
- Stock Management Hub - bulk stock edit, stock transfers, stock reports, reorder suggestions
- Customers Hub - customer management, trade accounts, invites, loyalty
- Communication Hub - WhatsApp, email, live chat, staff chat
- Reports Hub - sales reports, stock reports, staff performance, analytics
- Admin Settings - user management, security, notification settings
- Website Hub - homepage editor, navigation, categories, filters, collections, SEO

5. Supplier Management & Sync Hub

Supplier Products (Bulk Editor)

- Central product database - manage products from ALL suppliers in one place
- Bulk Category Editor - assign categories, filters, specs to multiple products at once
- Per-product scoping - assign different values (e.g. finish) to different products in one save
- Bulk price editing - cost price, list price, sale pricing
- Custom product labels with colour picker (Sale, Clearance, New, etc.)
- Series/collection renaming
- Product copy/duplicate functionality
- AI-powered description generator
- Product archiving and deletion with cascade cleanup
- Size and product filters for targeted bulk updates
- Quick search within value lists

Sync Hub (Automated Supplier Scraping)

- Automated web scrapers for suppliers: Ceramica Impex, Verona, Wallcano, Splendour
- Sync progress tracking with resume capability
- Force stop/reset controls
- Price Lock & Name Lock - protect manual edits from sync overwrites
- Validation warnings for data quality issues
- Sync staging area for review before publishing

Tier Pricing Engine

- Dynamic volume discount tiers (supports 2, 3, 4+ tiers)
- Per-product or global tier configuration
- Trade discount (%) - extra off for trade accounts
- Credit back rate (%) - credited back on trade orders
- Compound discount math: Sale + Volume + Trade discounts
- Supports decimal percentages (e.g. 7.5%)

Publishing Workflow

- "Add All to Database" - publish supplier products to the storefront
- Auto-syncs categories, specifications, filters, pricing to tiles collection
- Draft -> Published workflow with Super Admin approval
- Bulk price increase tool (e.g. 5% supplier price increase)

6. Stock Management

Stock Control

- Real-time stock levels across all showrooms and warehouse
- Bulk stock edit - update quantities for multiple products
- Stock import from spreadsheets
- Stock transfers between showrooms
- Delivery check-in - log received stock from suppliers
- Batch tracking for stock movements

Stock Intelligence

- Reorder suggestions - automatic alerts when stock falls below minimum
- Reorder points configuration per product
- Stock cost report - total stock value analysis
- Stocktake report - for physical stock counts
- To-order report - products that need reordering
- Stock allocation - reserve stock for specific orders

7. Sales & Documents

Document Types

- Invoices - full tax invoices with line items, VAT, payment terms
- Quotations - formal quotes with validity period and follow-up
- Cash Quotations - quick quotes for walk-in customers
- Proforma Invoices - pre-payment invoices for advance orders
- Credit Notes - credits for returns, damaged goods, price adjustments
- Refunds - full or partial refund processing

Document Features

- PDF generation and download
- Email documents directly to customers
- Document history with search and filters
- Order management with status tracking
- Abandoned cart recovery (website orders)

8. Customer & Trade Accounts

Customer Management

- Customer database with contact details and order history
- Customer registration (website or in-store)
- Customer invites - send registration links
- Customer account pages - order history, saved details

Trade Accounts

- Trade registration with business details
- Admin approval workflow for trade applications
- Trade pricing - automatic ex-VAT pricing for approved accounts
- Trade discount configuration per supplier or product group
- Trade list management
- WhatsApp welcome messages for new trade accounts

Loyalty Programme

- Loyalty points system
- Points earned on purchases
- Loyalty dashboard for tracking and management

9. Communication Hub

WhatsApp Integration

- Meta WhatsApp Cloud API integration
- Automated welcome messages for new trade registrations
- Manual send from Trade Accounts page (individual + bulk)
- Custom message composer
- Order status notifications via WhatsApp

Email

- Email composer for customer communications
- Email inbox management
- Automated email notifications for orders

Chat

- Live chat on the website for customer support
- Staff chat for internal communication
- SMS updates for order notifications

10. Reports & Analytics

Sales Reports

- Daily, weekly, monthly sales summaries
- Historical sales data analysis
- Sales by product, category, supplier
- Website sales dashboard

Stock Reports

- Stock valuation report
- Stocktake comparison report
- Low stock alerts and reorder suggestions

Staff & Performance

- Staff performance dashboard
- Sales by staff member
- Audit trail - track all admin actions

Website Analytics

- Conversion analytics - visitor to purchase funnel
- Storefront health monitoring
- Supplier health dashboard - track sync status and data quality

11. Website Management

Homepage

- Homepage content editor - banners, featured collections, promotions
- Homepage manager - layout and section ordering
- Welcome popup management

Navigation & Categories

- Navigation menu editor - customise the main menu structure
- Navigation structure manager - groups, categories, subcategories
- Categories manager - create, edit, delete categories per product group
- Independent categories per group (same name can exist in Tiles AND Flooring)

Filters & Specifications

- Filters manager - configure what customers can filter by (colour, size, finish, etc.)
- Specifications manager - product specs shown on detail pages
- Manage Options modal - assign specs/filters to product groups
- Bulk scope feature - assign multiple specs/filters to groups in one click

Collections

- Collection manager - customise collection pages
- Collection detail settings - hero images, descriptions, labels
- Collection mapping manager - control how products are grouped

Other Website Tools

- SEO management - sitemap, meta titles, descriptions
- Tile calculator settings
- Checkout settings
- Customer account settings
- Trade account settings
- Sample service content management
- Website preview tool
- Page maintenance mode

12. Security & User Management

User Roles

- Super Admin - full access to all features
- Staff - showroom access, EPOS, limited admin
- Trade Customer - trade pricing, order history
- Retail Customer - standard pricing, order history

Security Features

- JWT authentication with token management
- Staff PIN system for quick showroom login
- Audit trail - logs all admin actions with timestamps and user details
- Security settings - password policies, session management
- Role-based access control throughout the system

Tile Station

Tonbridge | Gravesend | Chingford

This document is auto-generated from the live system codebase.

For questions or updates, contact the development team.
